

1 Bias and Variance

1.1 Definition:

Error due to Bias:

The error due to bias is taken as the difference between the expected (or average) prediction of our model and the correct value which we are trying to predict. Of course, you only have one model so talking about expected or average prediction values might seem a little strange. However, imagine you could repeat the whole model building process more than once: each time you gather new data and run a new analysis creating a new model. Due to randomness in the underlying data sets, the resulting models will have a range of predictions. Bias measures how far off in general these models' predictions are from the correct value.

High Bias indicates model is fitting poorly to the training data, and one of the main reasons for this is we have considered wrong assumptions about the existing relationship of data (Fitting linear model to the quadratic relationships). When there is high bias, model is too general and it has created rules which are incapable to capture relationships between predictors (X) and response (Y).

Error due to Variance:

The error due to variance is taken as the variability of a model prediction for a given data point. Again, imagine you can repeat the entire model building process multiple times. The variance is how much the predictions for a given point vary between different realizations of the model.

High variance indicates that model is able to capture the underlying relationships very well, but it is incapable of producing similar performance on unseen dataset. Generalization ability of the model is too low.

When there is high variance, model is too specific to the training data and it has created rules which will not generalize well on the unseen dataset.

1.2 Mathematical Definition:

If we denote the variable we are trying to predict as Y and our covariates as X , we may assume that there is a relationship relating one to the other such as $Y=f(X)+\epsilon$ where the error term ϵ is normally distributed with a mean of zero like so $\epsilon \sim N(0, \sigma^2)$.

We may estimate a model $\hat{f}(X)$ of $f(X)$ using linear regressions or another modeling technique. In this case, the expected squared prediction error at a point x is:

$$\text{Err}(x) = E[(Y - \hat{f}(x))^2]$$

This error may then be decomposed into bias and variance components:

$$\text{Err}(x) = (E[\hat{f}(x)] - f(x))^2 + E[(\hat{f}(x) - E[\hat{f}(x)])^2] + \sigma^2$$

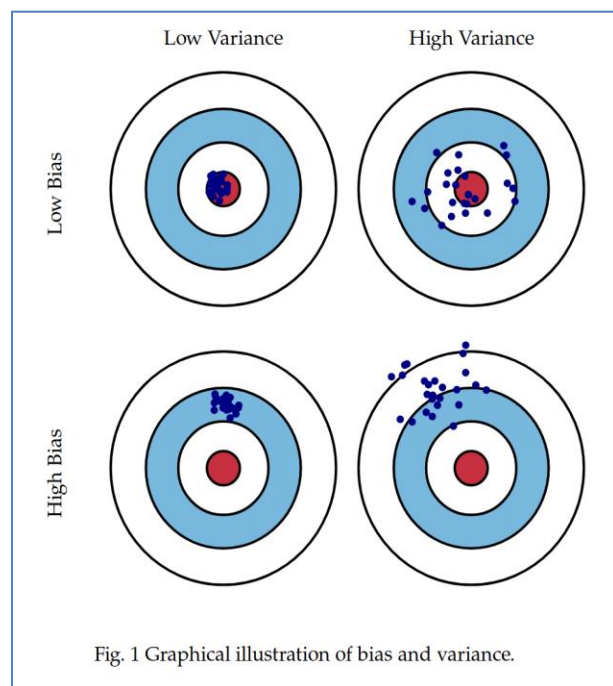
$$\text{Err}(x) = (\text{Bias}^2 + \text{Variance}) + \text{Irreducible Error}$$

That third term, irreducible error, is the noise term in the true relationship that cannot fundamentally be reduced by any model. Given the true model and infinite data to calibrate it, we should be able to reduce both the bias and variance terms to 0. However, in a world with imperfect models and finite data, there is a tradeoff between minimizing the bias and minimizing the variance.

1.3 Graphical Definition:

We can create a graphical visualization of bias and variance using a bulls-eye diagram. Imagine that the center of the target is a model that perfectly predicts the correct values. As we move away from the bulls-eye, our predictions get worse and worse. Imagine we can repeat our entire model building process to get a number of separate hits on the target. Each hit represents an individual realization of our model, given the chance variability in the training data we gather. Sometimes we will get a good distribution of training data so we predict very well and we are close to the bulls-eye, while sometimes our training data might be full of outliers or non-standard values resulting in poorer predictions. These different realizations result in a scatter of hits on the target.

We can plot four different cases representing combinations of both high and low bias and variance.



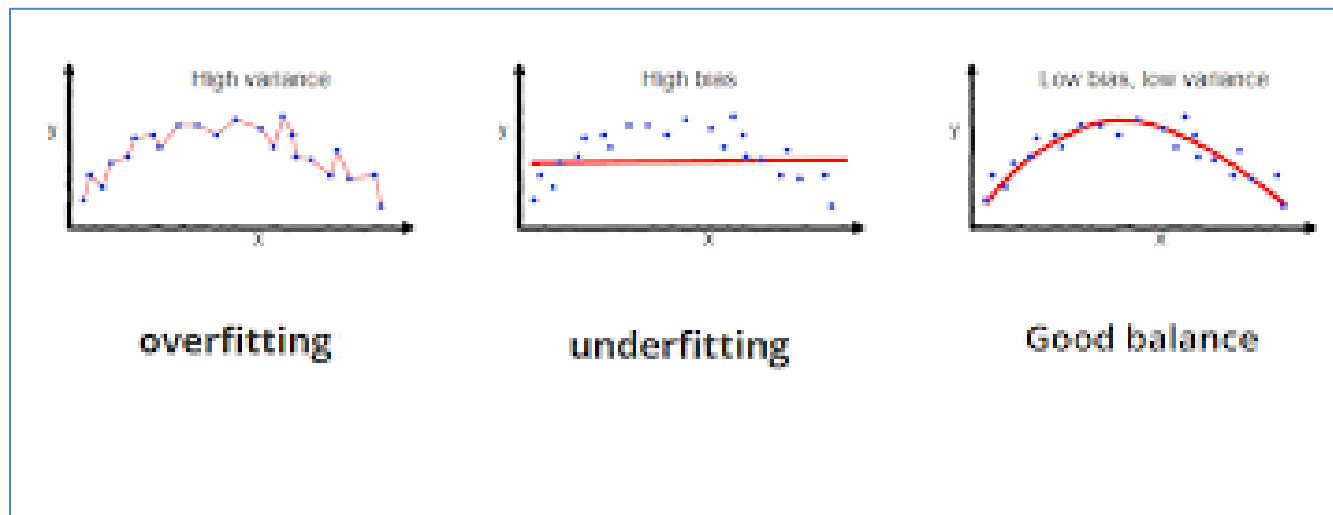
2. Overfitting and Underfitting

2.1 Overfitting:

When the model has low bias, it leads to the high variance problem. As mentioned earlier (High Variance), Overfitting happens when model learns underlying relationships well but lack the ability to produce similar performance on the unseen dataset. This can usually be verified using cross-validations or using a separate dataset to check the stability of the trained model. If validation error is significantly higher than training error then it shows that model is overfitted on the training dataset.

2.2 Underfitting:

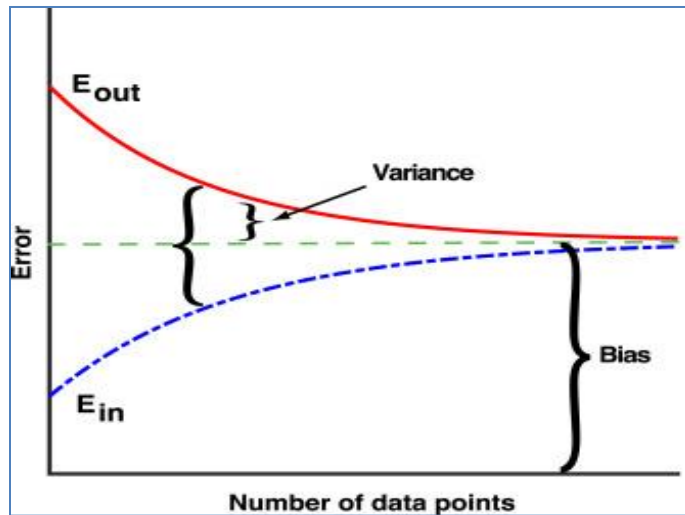
In contrast to Overfitting, when model has low variance, it leads to the problem of high bias. This is usually called as Underfitting problem where model does not have the capability to capture the underlying relationships between predictors and response.



3. Handling the problem of Overfitting and Underfitting

3.1 Overfitting:

Overfitting usually happens because the pattern recognized by the trained model is not able to generalize well on unseen dataset.



As can be observed from 3.1.1, when we have low sample size, training error (E_{in}) is low but there's a significant gap in training error (E_{in}) and validation error (E_{out}) which is extremely high comparatively. However, as we keep increasing the sample size, we observed that variance has reduced at the expense of high bias. **Hence, one of the solutions to overcome the problem of Overfitting is to increase the model training size.**

Fig. 3.1.1 Error vs. Sample Size

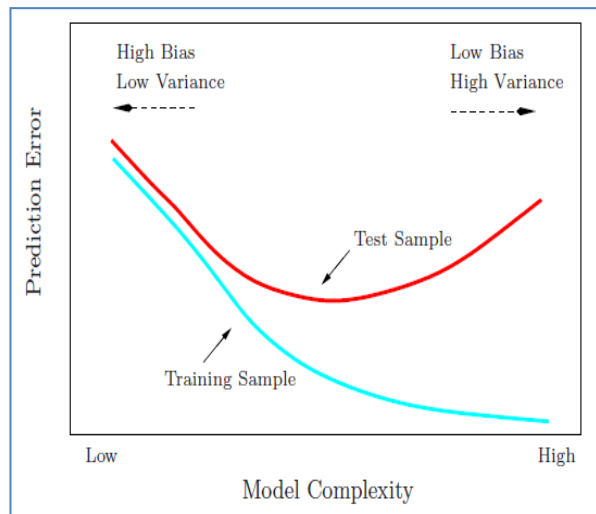


Fig. 3.1.2 Error vs. Model Complexity

Fig. 3.1.2 shows that complex models tend to overfit a lot compared to simple models. Prediction error (y-axis) reduces as the model complexity (x-axis) moves from low to high. However, test error shows somewhat a different behavior – reduces until a certain point and then it starts increasing as our model gets more and more complex. ***This shows that reducing the model complexity will help in reducing the high variance problem (Overfitting). Hence, removing the redundant variables from the model helps in overcoming the Overfitting problem.***

This can be done by removing variables based on the business understanding, using feature reduction techniques (like PCA, etc.) or based on low importance in model. ***Additionally, using regularization techniques like Ridge or Lasso regression, magnitude of the coefficients can be reduced and hence improving the model generalization ability.*** If model training and test datasets are showing different behavior for few variables then it might be a good idea to use stratified sampling and reduce the Overfitting problem. However, we need to ensure that model performs well on the holdout sample as well.

3.2 Overcoming the problem of Underfitting

As seen in fig. 3.1.1, adding more observations reduces the variance. However, it's a misconception that this helps in Underfitting problem. When the model is under-fitted, it means that there is some missing information in model which is affecting the accuracy of the model. Now, if we add that information by the means of additional variables, we will be able to reduce the problem of high bias. Hence, getting more information in the form of additional features (either created from existing or using external sources), will help in reducing the bias of the model and eventually solving the problem of Underfitting.